1. Print messages

```
fun main() {
    println("Use the val keyword when the value doesn't change.")
    println("Use the var keyword when the value can change.")
    println("When you define a function, you define the parameters that can be passed to it.")
    println("When you call a function, you pass arguments for the parameters.")
}
```

---

2. Fix compile error

```
fun main() {
    println("New chat message from a friend")
}
```

---

3. String templates

```
fun main() {
    val discountPercentage: Int = 20
    val item = "Google Chromecast"
    val offer = "Sale - Up to $discountPercentage% discount on $item! Hurry up!"
    println(offer)
}
```

---

4. String concatenation

```
fun main() {
    val numberOfAdults = 20
    val numberOfKids = 30
    val total = numberOfAdults + numberOfKids
    println("The total party size is: $total")
}
```

---

5. Message formatting

```
fun main() {
    val baseSalary = 5000
    val bonusAmount = 1000
    val totalSalary = baseSalary + bonusAmount
    println("Congratulations for your bonus! You will receive a total of $totalSalary (additional bonus).")
}
```

---

6. Implement basic math operations

```
fun main() {
    val firstNumber = 10
```

```
    val secondNumber = 5
    val thirdNumber = 8

    val result = add(firstNumber, secondNumber)
    val anotherResult = add(firstNumber, thirdNumber)

    println("$firstNumber + $secondNumber = $result")
    println("$firstNumber + $thirdNumber = $anotherResult")

    println("$firstNumber - $secondNumber = ${subtract(firstNumber,
secondNumber)}")
}

fun add(a: Int, b: Int): Int = a + b
fun subtract(a: Int, b: Int): Int = a - b
```

7. Default parameters

```
fun main() {
    val firstUserEmailId = "user_one@gmail.com"
    println(displayAlertMessage(emailId = firstUserEmailId))
    println()

    val secondUserOperatingSystem = "Windows"
    val secondUserEmailId = "user_two@gmail.com"
    println(displayAlertMessage(secondUserOperatingSystem,
secondUserEmailId))
    println()

    val thirdUserOperatingSystem = "Mac OS"
    val thirdUserEmailId = "user_three@gmail.com"
    println(displayAlertMessage(thirdUserOperatingSystem,
thirdUserEmailId))
    println()
}

fun displayAlertMessage(operatingSystem: String = "Unknown OS", emailId:
String): String {
    return "There's a new sign-in request on $operatingSystem for your
Google Account $emailId."
}
```

8. Pedometer

```
fun main() {
    val stepsTaken = 4000
    val caloriesBurned = calculateCaloriesFromSteps(stepsTaken)
    println("Walking $stepsTaken steps burns $caloriesBurned calories")
}

fun calculateCaloriesFromSteps(steps: Int): Double {
    val caloriesPerStep = 0.04
    return steps * caloriesPerStep
}
```

9. Compare two numbers

```
fun main() {
    println(hasMoreScreenTimeToday(timeSpentToday = 300, timeSpentYesterday
= 250))
    println(hasMoreScreenTimeToday(timeSpentToday = 300, timeSpentYesterday
= 300))
    println(hasMoreScreenTimeToday(timeSpentToday = 200, timeSpentYesterday
= 220))
}

fun hasMoreScreenTimeToday(timeSpentToday: Int, timeSpentYesterday: Int):
Boolean {
    return timeSpentToday > timeSpentYesterday
}
```

---

10. Move duplicate code into a function

```
fun main() {
    printWeather("Ankara", 27, 31, 82)
    printWeather("Tokyo", 32, 36, 10)
    printWeather("Cape Town", 59, 64, 2)
    printWeather("Guatemala City", 50, 55, 7)
}

fun printWeather(city: String, lowTemp: Int, highTemp: Int, chanceOfRain:
Int) {
    println("City: $city")
    println("Low temperature: $lowTemp, High temperature: $highTemp")
    println("Chance of rain: $chanceOfRain%")
    println()
}
```